

Enterprise Cloud Migration Strategy

1. Introduction

The organization intends to migrate its existing on-premises infrastructure to a hybrid cloud architecture to improve scalability, operational resilience, resource utilization, and cost efficiency. The proposed migration strategy involves the progressive modernization of legacy workloads while maintaining interoperability with critical systems that cannot immediately be migrated.

2. Current Infrastructure

The existing infrastructure consists of physical servers, virtual machines, relational database systems, file servers, internal applications, and enterprise networking components. A significant proportion of these workloads operate on aging infrastructure, resulting in increased maintenance requirements and limited scalability.

The current architecture also creates dependencies between application services and infrastructure components, thereby increasing the complexity associated with system upgrades and maintenance activities.

3. Proposed Architecture

The proposed hybrid cloud architecture will distribute workloads between on-premises infrastructure and cloud-based services. Workloads requiring low latency, regulatory compliance, or direct integration with internal systems will remain on-premises, while scalable and less infrastructure-dependent workloads will be migrated to the cloud.

Cloud resources will utilize automated provisioning, centralized monitoring, identity and access management, and elastic resource allocation.

4. Migration Approach

The migration will be executed incrementally through the following phases:

Phase 1 – Assessment

All applications, databases, dependencies, infrastructure requirements, and security considerations will be documented. Each workload will be evaluated according to business criticality, technical complexity, migration feasibility, and operational risk.

Phase 2 – Pilot Migration

A limited number of non-critical workloads will be migrated to validate network connectivity, security controls, application compatibility, monitoring capabilities, and performance characteristics.

Phase 3 – Production Migration

Following successful validation of the pilot environment, critical workloads will be migrated according to predefined migration schedules and rollback procedures.

Phase 4 – Optimization

After migration, cloud resources will be continuously monitored and optimized. Resource utilization, performance, availability, and operational expenditure will be evaluated periodically.

5. Security Considerations

The migration must implement identity-based access control, encryption of data at rest and in transit, network segmentation, centralized logging, vulnerability management, and continuous security monitoring.

Access privileges should follow the principle of least privilege to minimize unauthorized access and potential security exposure.

6. Risks

Potential risks include application incompatibility, data loss, service interruption, unexpected cloud expenditure, security vulnerabilities, and inadequate employee training.

To mitigate these risks, the organization should maintain backups, implement rollback procedures, conduct security assessments, establish cost-monitoring mechanisms, and provide appropriate technical training.

7. Expected Benefits

The migration is expected to improve infrastructure scalability, reduce hardware dependency, increase operational flexibility, improve disaster recovery capabilities, and enable more efficient utilization of computing resources.